# Chan Hur

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## Research Interests

Multimodal learning and representation learning, with a focus on vision-language models, text–video retrieval, GUI agents, compositional image retrieval, video question answering, and applications of large language and multimodal models. Research interests include cross-modal alignment, generative caption utilization, bridging semantic gaps between modalities, and domain-specific AI for medical applications.

## Research Experience

**Researcher, ETRI** 2026.02–Present
Digital Convergence Research Department
Electronics and Telecommunications Research Institute
Research on vision-language models, multimodal retrieval, and GUI agents.

## Education

**Ph.D. in Computer Science and Engineering** 2021.03–2026.02
Kyungpook National University, Daegu, South Korea
Dissertation: *Advancing Text-Video Retrieval with Frame-Wise Generative Information*
Advisor: Prof. Hyeyoung Park

**M.S. in Computer Science and Engineering** 2019.03–2021.02
Kyungpook National University, Daegu, South Korea
Thesis: *Zero-Shot Image Recognition Model Using Image Description*
Advisor: Prof. Hyeyoung Park

**B.S. in Mechanical Engineering** 2014.03–2019.02
Kyungpook National University, Daegu, South Korea

## Publications

### International Conferences

**Surface-Attribute Text Conditioning for RGB-to-HSI Reconstruction**
**Chan Hur**, Woonggyu Min, Ho-Youl Jung, Soo Jun Park, Myungeun Lim
*Asian Conference on Computer Vision (ACCV), 2026. Accepted.*

**Prescription Orders as a Prediction-Time Signal for Forward Acute Kidney Injury**
Hyung Wook Noh, **Chan Hur**, Jeonghyun Kim
*IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2026. Accepted.*
*[OpenReview]*

**PopupBench: Diagnosing Popup Blindness in Vision–Language GUI Agents**
**Chan Hur***, Youngjae Lee*, Hyeyoung Park, Hyung Wook Noh, Youngwan Lee
*NeurIPS 2026 Workshop on Interpreting Agent Behavior (IAB). Accepted.*

**Semantic Hardness Is Not Visual Hardness: Sign-Aware Hard Negative Mining for Sign Language Retrieval**
Junmyeong Lee, **Chan Hur**, ChangSu Choi, Sukmin Cho, Fitsum Gaim, Eui Jun Hwang, Hoyun Song, KyungTae Lim
*Annual Meeting of the Association for Computational Linguistics (ACL), 2026.*

**ReFineVQA: Iterative Refinement of Video Description via Feedback Generation for Video Question Answering**
Jeongwan Shin*, **Chan Hur***, Seongmin Cho, Jae-Ho Choi, Hyeyoung Park
*Winter Conference on Applications of Computer Vision (WACV), 2026.*

**Narrating the Video: Boosting Text-Video Retrieval via Comprehensive Utilization of Frame-Level Captions**
**Chan Hur***, Jeong-hun Hong*, Dong-hun Lee, Dabin Kang, Semin Myeong, Sang-hyo Park, Hyeyoung Park
*Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [arXiv]*

**Local Feature Aggregation with Attribute Anchors for Text-guided Image-Text Retrieval**
**Chan Hur**, Hyeyoung Park
*International Conference on Artificial Intelligence in Information and Communication (ICAIIC), 2025. [IEEE]*

**Automatic Image Recommendation for Economic Topics using Visual and Semantic Information**
**Chan Hur**, Changhun Hyun, Hyeyoung Park
*International Conference on Semantic Computing (ICSC), 2020. [IEEE]*

#### International Journals

**EENet: Embedding Enhancement Network for Compositional Image-Text Retrieval using Generated Text**
**Chan Hur**, Hyeyoung Park
*Multimedia Tools and Applications, 2023. SCIE (Q1). [DOI]*

**Zero-shot Image Classification using Captioning-based Rectified Embedding Vector**
**Chan Hur**, Hyeyoung Park
*Applied Sciences, 2023. SCIE (Q2). [DOI]*

**An Image Selection Framework for Automatic Report Generation**
Changhun Hyun, **Chan Hur**, Hyeyoung Park
*Multimedia Tools and Applications, 2022. SCIE (Q1). [DOI]*

#### Preprints & Under Review

**PopupBench: Revealing Popup Blindness in Vision-Language GUI Agents**
**Chan Hur**, Youngjae Lee, Hyeyoung Park, Youngwan Lee
*Under review, October 2026.*

**TraceCIR: Executable Intent Trajectories for Stateful Language Grounding in Multi-Turn Image Retrieval**
**Chan Hur**, Minkyung Cho, Jun Seo Kim, Won Jun Oh, Hyung Wook Noh, KyungTae Lim
*Under review, October 2026.*

**MELON: A Large-Scale Dataset for Multi-Event Text-to-Long-Video Retrieval**
**Chan Hur***, SeungWoo Song*, Jeong-hun Hong*, Won Jun Oh, Hyeyoung Park, KyungTae Lim
*arXiv preprint arXiv:2609.01654, 2026. [arXiv]*

#### Domestic Publications

**Improving text-video retrieval performance using video features combined with generation-based caption information**
**Chan Hur**, Dong-hun Lee, Sang-hyo Park, Hyeyoung Park
*KIISE Transactions on Computing Practices, KCI, 2024.*

**Feature Vector Combination on Joint Embedding Space using Captions for Text Video Retrieval**
**Chan Hur**, Dong-hun Lee, Sang-hyo Park, Hyeyoung Park
*Korea Computer Congress (KCC), 2023.* ***Best Paper Award.***

**Deep learning-based Lightweight Model for Carbon-Reduction Activity Image Classification**
Hojun Song, Yunhee Koo, Sehyun Park, Jeoungwon Cha, Youngjae Lee, **Chan Hur**, Hyeyoung Park
*Journal of Embedded Systems and Applications, KCI, 2022.*

**A Study on the Alternative Method of Video Characteristics Using Captioning in Text-Video Retrieval Model**
Dong-hun Lee, **Chan Hur**, Hyeyoung Park, Sang-hyo Park
*Journal of Embedded Systems and Applications, KCI, 2022.*

**A study on Video-text Retrieval model using Caption Information**
Dong-hun Lee, **Chan Hur**, Hyeyoung Park, Sang-hyo Park
*Korea Artificial Intelligence Conference, 2022.*

## Patents

**전자 장치 및 그의 이미지를 추천하는 방법** 2025.05
Registered (등록).

**언어 생성모델을 이용한 대화형 이미지 추천 시스템** 2023.06
Patent application (출원).

**텍스트 기반 비디오 검색 장치 및 그 비디오 검색 방법** 2025.05
Registered (등록).

## Awards

**Best Graduate Ph.D. Student Award** 2026
Department of Computer Science and Engineering, Kyungpook National University

**Best Paper Award** 2024
LSTM 기반 온실 환경 예측 및 제어 시스템 최적화
ICT 대학생 논문 경진대회

**Best Paper Award** 2023
Feature Vector Combination on Joint Embedding Space using Captions for Text Video Retrieval
Korea Computer Congress (KCC)

## Teaching

**Deep Learning and Machine Learning with Python** 2023–2025
Kyungpook National University IT Education Center

**Introduction of Generative AI and Applications** 2024–2025
Kyungpook National University IT Education Center

**Machine Learning** 2023–2025
Korea National Open University

## Skills

**Deep Learning Frameworks:** PyTorch, TensorFlow
**Languages:** Korean (native), English (fluent), Japanese (intermediate)